

Continue â€” Project Status Snapshot

Revision: 16 **Last modified:** 2026-06-10T05:12:40Z **Session:** 2026-06-10 (Session 10 â€” BobaLink browser extension: discovery + 9-phase plan + Phase 1 + backend BE-1/BE-2 + env-gated token security fix) **Last commit:** 192b945 (env-gated BOBA_API_TOKEN on the three :7187 download-write endpoints) **Branch:** main **Working tree:** CLEAN (after the doc commit that carries this update)

Send continue to pick up exactly where we left off. This file is the single source of truth for session handoff (Â§12.10 / Â§11.4.131).

CURRENT STATE â€” Session 10 (BobaLink browser extension)

New major feature in flight: the BobaLink browser extension. Plan + discovery + Phase 1 foundation + backend BE-1/BE-2 + a token-auth security fix landed. The Python unit suite stays FULLY GREEN at 4149 passed.

What is BobaLink

A WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs on any page and forwards them to the running Boba merge service on port **7187**. Backend contract: POST http://<host>:7187/api/v1/download. The full plan, discovery analysis, and traceability live under docs/browser_extension/ â€” master plan IMPLEMENTATION_PLAN.md (9 phases), analysis artifacts _analysis/01â€”06, planning artifacts _plan/Aâ€”F, and the 245-item traceability matrix _plan/C-traceability-matrix.md (240 v1 / 5 v2, proving nothing is skipped per Â§11.4.118).

What landed (Session 10 â€” BobaLink)

- b2356ae â€” BobaLink master implementation plan + discovery/planning artifacts (docs/browser_extension/); also bumped HelixQA submodule bcac236 â†’ 4d2dcb2.
- 33a9815 â€” extension **Phase 1 foundation** scaffolded (extension/ â€” WXT config, TS, shared libs/types under extension/src/); **71 anti-bluff Vitest unit tests** (vitest 71 passed, eslint clean â€” crypto imports the REAL module, not an inline copy).
- d46bffb â€” backend **BE-1** (CORS for extension origins) + **BE-2** (raw .torrent upload endpoint) added to the Python :7187 API.
- 284d1c4 â€” bumped HelixQA submodule 4d2dcb2 â†’ bca3b36; new Challenge bank submodules/helixqa/banks/boba-bobalink.yaml (6 cases).
- 192b945 â€” **security fix:** env-gated shared-secret BOBA_API_TOKEN enforced on the three :7187 download-write endpoints (when the env var is set, requests must carry a matching X-Boba-Token / Bearer token; behaviour unchanged when unset). Verified by **15 token-auth tests** (tests/unit/api_layer/test_download_token_auth.py).

Suite status

- Full Python unit suite: **4149 passed, 6 warnings in 358.95s** â€” `pytest tests/unit/ --import-mode=importlib` (this = 4134 + the 15 new token-auth tests). Latest evidence: `qa-results/tokenauth_fullsuite_*.log` (tail shows 4149 passed).
- Extension Vitest: **71 passed** (foundation libs; commit 33a9815).

Current phase + next steps

- **Phase 2 (core detection & parsing engine) IN PROGRESS** â€” porting/refactoring the parsers (`bencode.ts`, `magnet.ts`, wiring the `dead-torrent-file.ts` SHA-1 infohash) and scanners (`link-scanner/text-scanner/content-scanner`, merging the divergent `site-db.ts` selector tables, `MutationObserver` debounce, `Shadow-DOM/iframe` handling) into `extension/src/`, plus infohash dedup.
- **Next:** finish Phase 2 scanner integrators + content/dedup, then Phases 3â€”9 (extension shell â†’ Boba backend client â†’ notifications/queue â†’ secure token model â†’ packaging â†’ release) per `docs/browser_extension/IMPLEMENTATION_PLAN.md`. Live working state also tracked in `.remember/remember.md`.

CURRENT STATE â€” Session 10 (2026-06-10, subagent-driven loop)

Suite stays **FULLY GREEN**. This session was **hardening + verification, subagent-driven** (Â§11.4.70), every change code-reviewed before commit (Â§11.4.125/Â§11.4.134).

What landed (Session 10)

- `test_credential_env_wiring.py` **isolation fix** â€” resolved the Session-9 â€œknown low-priority follow-upâ€” below. A `merge_service` fake-namespace stub was installed at module-collection time with no teardown, so `conftest`â€™s `_isolate_download_proxy_modules` captured-and-restored the *fake* stub into siblings in some run orders. Fixed by moving stub install into a `search_mod` fixture that snapshots/restores the 3 injected keys (`merge_service`, `.search`, `.retry`) in finally. **Evidence:** RED reproduction (the 5 credential tests + a temporary leak-probe) went 1 failed â†’ all green; the fileâ€™s own **5 tests pass** standalone; full unit suite **4121 passed** — seeds default/42/31337 (Session-10 run). Code-review **GO** (proved negation; no weakened assertions; ruff clean).
- `docs/qa/BOB-008/operator_runbook.md` â€” copy-pasteable operator unblock procedure for BOB-008 (cookie path preferred; CAPTCHA path documented with its `cap_sid/cap_code_field` friction), every endpoint claim cited to `auth.py:line` (Â§11.4.83/Â§11.4.99).
- **Latent-leak discovery sweep** (Â§11.4.118) â€” audited all of `tests/` for the module-level fake-stub-no-teardown pattern. Finding: the pattern is a **structural no-op wherever the test dir has an `__init__.py`** (real package pre-loads â†’ `setdefault` never installs the fake), so the ~26 `tests/unit/merge_service/` matches and `test_tracker_stats.py` are **inert** (no RED reproduces; strict TDD â†’ no fix). The single genuinely-uncovered root is `pirateiro` (see follow-up).
- **Constitution inheritance verified** (Â§11.4.32/Â§11.4.35) â€” PASS: pointer present in both layers, propagated anchors present in canonical source, no conflict markers, `pre_build_verification.sh` 18/18 green.

- **pirateiro test-isolation fix** **BOB-063 (operator-approved, implemented)** `tests/unit/test_plugin_pirateiro.py` injected `sys.modules['pirateiro']` at module scope with no teardown; `pirateiro` was the one root uncovered by `conftest`'s isolation, so it leaked into later tests. The naive `add to _POLLUTING_ROOTS` does NOT work (the leak is already inside each per-test snapshot); the real fix caches+re-registers+purges the stub per unit test. **Evidence (Â§11.4.115):** RED 1 failed, 44 passed **GREEN** 45 passed; negation proof (disable the purge **re-fails**); full suite **4122 passed** — **seeds default/42**. Standing regression guard `tests/unit/test_pirateiro_isolation_guard.py` (Â§11.4.135). QA at `docs/qa/BOB-063/evidence.md`. **Closed BOB-063** (Task **Completed**).
- **Constitution submodule advanced** `60e2d66` **26368b** (Â§11.4.26, **clean fast-forward, Â§11.4.113-safe**) pulled `Â§11.4.140 v2` (BACKGROUND action), `Â§11.4.142` (universal code-review **already** actively enforced this session; added to `CLAUDE.md` propagated clauses), `Â§11.4.143` (video-streaming real-user-journey **latent/N-A** to this torrent-proxy project). Cascade gate **18/18 green** at the new pin.

Open queue (Issues.md)

- **BOB-008** **RuTracker CAPTCHA** **OPERATOR-BLOCKED**. Unblock procedure documented at `docs/qa/BOB-008/operator_runbook.md` (preferred: paste `bb_session` via `POST /api/v1/auth/rutacker/cookie-login`).
- Everything else closed (DB **iZMD** in sync, **63 items**, `bin/workable-items` validate OK).

Known low-priority follow-up

- ~~`pirateiro test-isolation defense-in-depth`~~ **RESOLVED this session** (BOB-063, see above).
- ~~`test_credential_env_wiring / env_loader stub-teardown quirk`~~ **RESOLVED this session** (see above).

CURRENT STATE **Session 9 (2026-06-10, overnight autonomous loop)**

The unit suite is now **FULLY GREEN** and **DETERMINISTIC**.

Metric	Value
Unit tests	4121 passed, 0 failed <code>pytest tests/unit/ --import-mode=importlib (5m24s)</code>
Determinism (Â§11.4.50)	top-level scope 3150 passed identical across <code>--randomly-seed 7/42/100/31337/12345</code>
Commits this session	<code>6e15a8d</code> (crash guards + <code>async/loop</code> hangs), <code>6230865</code> (test-pollution + network timeouts) both pushed to all upstreams
Code review	GO (zero findings/warnings) after <code>Â§11.4.134</code> iterate-until-GO

What landed (Session 9)

- **8 product fixes** (6e15a8d): degenerate-input crash guards for tokyotoshokan/kickass/yts/piratebay; enricher full-suite-hang fix (aiohttp.ClientTimeout—6); kickass/bitsearch/torrentgalaxy unbounded-loop caps (MAX_PAGES=50); mutation-scanner .venv scope fix; +7 download_proxy tests. Closed **BOB-060**.
- **Test-suite stabilization** (6230865): eliminated all Â§11.4.50 order-dependent pollution â€” tests/conftest.py _CORRECT_MS_PATH repo-root fix (the dominant bug: corrupted merge_service.__path__ â†’ broke 11 tests incl.Â all scheduler_api), extended _POLLUTING_ROOTS, + per-test socket.socket & os.environ snapshot/restore. Network-I/O timeout hardening (search.py/routes.py/helpers.py/eztv.py). Closed **BOB-061**, **BOB-062**.

Open queue (Issues.md)

- **BOB-008** â€” RuTracker CAPTCHA â€” **OPERATOR-BLOCKED**: needs you to complete the CAPTCHA at /api/v1/auth/rutracker/captcha + /login, OR paste a fresh bb_session cookie via /auth/rutracker/cookie-login. Cannot be solved autonomously.
- Everything else closed (DBâ†’i,ŽMD in sync, 62 items, bin/workable-items validate OK).

How to run the suite (both work now)

- Monolithic: .venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib â†’ 4121 passed.
- Per-scope (faster, parallelizable): tests/unit/merge_service/ (801) Â tests/unit/api_layer/ (170) Â tests/unit/*.py top-level (3150).
- Always pass --import-mode=importlib (the merge_service.deduplicator lazy import needs it; a bare per-file run errors).

Known low-priority follow-up (NOT blocking â€” suite is green)

- A residual test_credential_env_wiring/env_loader sys.modules-stub ordering quirk surfaces only in narrow isolated file-combos; it does NOT manifest in the full top-level scope (5 seeds green) because the new env/sys.modules isolation covers it there. Tighten the env_loader stub teardown if it ever resurfaces.

Session 8 Summary

Metric	Value
Test files	47 test modules (222 test_*.py files total across all suites)
Test cases	4,074+ passed
Coverage (unit)	88%+ (gate: 49%)
Bugs fixed	21 (BOB-001 through BOB-025, minus BOB-008)
Submodules aligned	constitution, challenges, containers, helixqa, jactett

Metric

Ruff / Mypy

Value

pre-existing warnings only, 0 new

Bug Tracker

21 bugs fixed across Session 8: - 10 B-substring parsing fixes - 3 import re missing fixes - 2 comma-size parsing fixes - 2 crash guards - 1 bt4g hang fix - 1 dedup circular import fix - 1 env_loader flaky test fix - 1 pirateiro full test suite (44/44)

1 remaining: BOB-008 (RuTracker CAPTCHA) â€” operator-blocked, needs manual cookie paste.

Test File Inventory

tests/unit/merge_service/ (31 files)

- test_cors_config.py
- test_dead_tracker_bucket.py
- test_deadline_tunable.py
- test_deduplicator.py
- test_deduplicator_edge.py
- test_diag_no_stale_leakage.py
- test_edge_case_challenges.py
- test_enricher.py
- test_enricher_edge.py
- test_hooks.py
- test_html_parsers.py
- test_jackett_autoconfig.py
- test_nnmclub_session_login.py
- test_private_tracker_html_fixtures.py
- test_public_plugin_harness.py
- test_public_plugin_harness_broad.py
- test_public_tracker_capture.py
- test_public_tracker_subprocess_timeout.py
- test_quality_detection.py
- test_scheduler.py
- test_scheduler_coverage.py
- test_search_concurrency.py
- test_search_coverage.py
- test_search_deep_coverage.py
- test_search_error_paths.py
- test_session_encryption.py
- test_theme_endpoint.py
- test_tracker_stats.py
- test_ttl_caches.py
- test_validator.py
- test_validator_coverage.py

tests/unit/api_layer/ (14 files)

- test_auth_coverage_extra.py
- test_concurrent_writers.py
- test_cors_config.py
- test_hooks_coverage.py
- test_hooks_endpoints.py

test_hooks_remaining.py
test_nnmclub_auth_endpoints.py
test_routes_coverage.py
test_sse_disconnect.py
test_sse_token_auth.py
test_theme_state_coverage.py
test_theme_stream.py
test_tracker_stats_sse.py

tests/unit/ (55 files, infrastructure + plugin test

test_auth.py, test_auth_coverage.py, test_auth_models.py
test_config.py, test_main.py, test_dashboard.py
test_content_type_refinement.py
test_credential_env_wiring.py
test_download_proxy_coverage.py, test_download_proxy_deep.py
test_download_merged.py
test_env_loader.py
test_freeleech.py
test_graceful_shutdown.py
test_helpers.py
test_log_filter.py
test_merge_trackers.py
test_novaprinter.py
test_private_tracker_search.py
test_public_tracker_subprocess.py
test_retry_policy.py
test_routes.py, test_routes_coverage.py
test_scheduler_api.py
test_sorting_weights.py
test_sse_disconnect.py
test_streaming.py
test_theme_injector.py, test_theme_wiring.py
test_webui_theme_injector.py
test_ui_module.py, test_ui_sorting.py
test_plugin_*.py

(39 plugin modules: eztv, piratebay, s
limetorrents, torlock, gamestorrents,
anilibra, torrentgalaxy, yts, rutor,
snowfl, torrentdownload, linuxtracker
nnmclub, rutracker, megapeer, jackett
one337x, extratorrent, torrentfunk, t
therarbg, academictorrents, ali213, y
glotorrents, pctorrent, rockbox, bitr
torrentscsv, xfsb, yihua, pirateiro,

test*_guards.py, test*_deep.py (crash guards + deep coverage variants
test_ci_infra.py, test_ci_workflows.py
test_community_plugins_compile.py
test_courses_scaffold.py, test_course_scripts_lint.py
test_docs_presence.py
test_frontend_spec_coverage.py
test_install_plugin_json_config.py
test_jackett_integration.py, test_jackett_plugin_pool.py
test_nnmclub_config_selfheal.py, test_nnmclub_plugin_login.py
test_no_runtime_service_skips.py
test_openapi_frozen.py

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test_page_title.py, test_footer_restored.py
test_palette_catalog.py, test_palette_catalog_python_mirror.py
test_quality_compose.py, test_quality_detection.py
test_readme_landing_page.py
test_runtime_requirements_includes_new_deps.py
test_scan_script_non_interactive.py
test_scanner_configs.py
test_service_fixtures.py
test_shadow_tokens.py
test_socks_udp.py
test_start_sh_copy_plugins_framework.py
test_toolchain_config.py
test_tracker_validator.py
test_website_config.py
test_branding_assets.py
test_architecture_diagrams.py
test_build_releases_non_interactive.py
tests/benchmark/ (3)
tests/chaos/ (2)
tests/concurrency/ (3)
tests/contract/ (5)
tests/docs/ (3)
tests/e2e/ (7)
tests/integration/ (19)
tests/memory/ (1)
tests/observability/ (2)
tests/performance/ (1)
tests/property/ (2)
tests/security/ (10)
tests/stress/ (2)

```

Commits (latest 10+)

```

c8254ee fix: 21 test failures, pirateiro 44/44, coverage gate 88%, dedup c
ccd0dc9 fix: env_loader flaky test + coverage baseline 88.14% + minor test
3d1819c docs: finalize Session 8 â€” Fixed.md Rev 12, 59 items, 18 bugs, 4
f924841 test: wave 8 â€” btsow/torrentscsv/xfsb/yihua (170 tests) + bt4g
6773203 test: wave 7 â€” yourbittorrent/glotorrents/pctorrent/rockbox/bitr
f191401 test: wave 6 â€” torrentfunk/torrentproject/therarbg/academicorre
1446cc5 test: wave 5 â€” audiobookbay/one337x/extratorrent (155 tests) + 3
fd543b0 docs: update CONTINUATION.md for Session 8 (4 waves, 23 plugins)
aae4069 test: wave 4 â€” kinozal/nmclub/rutracker/megapeer/jackett + mega
e3b7acb test: wave 3 â€” rutor/tokyotoshokan/snowfl/torrentdownload/linux

```

Coverage Snapshot

Module	Coverage
api/init.py	98%
api/auth.py	91%
api/routes.py	95%

Module	Coverage
api/streaming.py	99%
main.py	94%
deduplicator.py	94%
enricher.py	100%
hooks.py	95%
jackett_autoconfig.py	99%
scheduler.py	93%
search.py	~90%
validator.py	92%
theme_injector.py	99%
env_loader.py	100%
download_proxy.py	~95%
TOTAL (unit)	88%+

Known Issues

1. **BOB-008**: RuTracker CAPTCHA “ operator-blocked (needs manual cookie paste)
 2. Go backend is a skeleton (documented in AGENTS.md)
 3. Containers may be down on session start “ bash start.sh first
 4. macOS + podman network_mode: host does NOT forward ports “ ensure-macos-tunnel.sh handles this
-

Quick-Start

```
# Pre-build gate
bash scripts/pre_build_verification.sh
```

```
# Unit tests
python3 -m pytest tests/unit/ -q --import-mode=importlib
```

```
# Coverage
python3 -m pytest tests/unit/ --cov=download-proxy/src --cov=plugins --cov=
```

```
# Lint + typecheck
ruff check . && mypy download-proxy/src/
```

```
# Frontend
cd frontend && npx vitest run
```

```
# Containers
bash start.sh && bash stop.sh
```

Architecture

Port	Service	Tech
7185	qBittorrent WebUI	LinuxServer (admin/admin)
7186	Download Proxy	Python HTTP
7187	Merge Search Service	FastAPI + Angular SPA
7188	WebUI Bridge	Python (host process)
7189	boba-jackett	Go/Gin
9117	Jackett	LinuxServer (auto-configured)